Background and Business Understanding

- Global mobile messaging >190T msgs by 2026
- 56% growth to 2030

- SA: phishing capital – 52% of cyber threats (ESET)
- 3-12 spam SMSs /week on new SIMs (Steyn- MBB)

- Operators face $80B global fraud cost ; reputation & financial harm

- POPIA: central raw SMS prohibited – fines up to R10M
- Need: high spam recall + low false positives + privacy-aware + scalable

Data Science Trajectory

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Business Understanding

- SA phishing + POPIA → need privacy-preserving spam detector
- Needs, spam recall priority

Data Understanding

- Dataset with 5,700 msgs, 12.3% spam, 84% duplicates, strong class separation

Data Preparation

- Keep letters +digits (amounts/short codes)
- no stop-words, no stemming → vocab ~330

Modelling

- Central: custom NB with TF-IDF (unigrams ± bigrams);
- Federated: horizontal NB on BoW counts + local DP (ε=0.05,1.0)

Evaluation

- Stratified group CV (group = raw hash) + spam-centric metrics (precision, recall, F1)

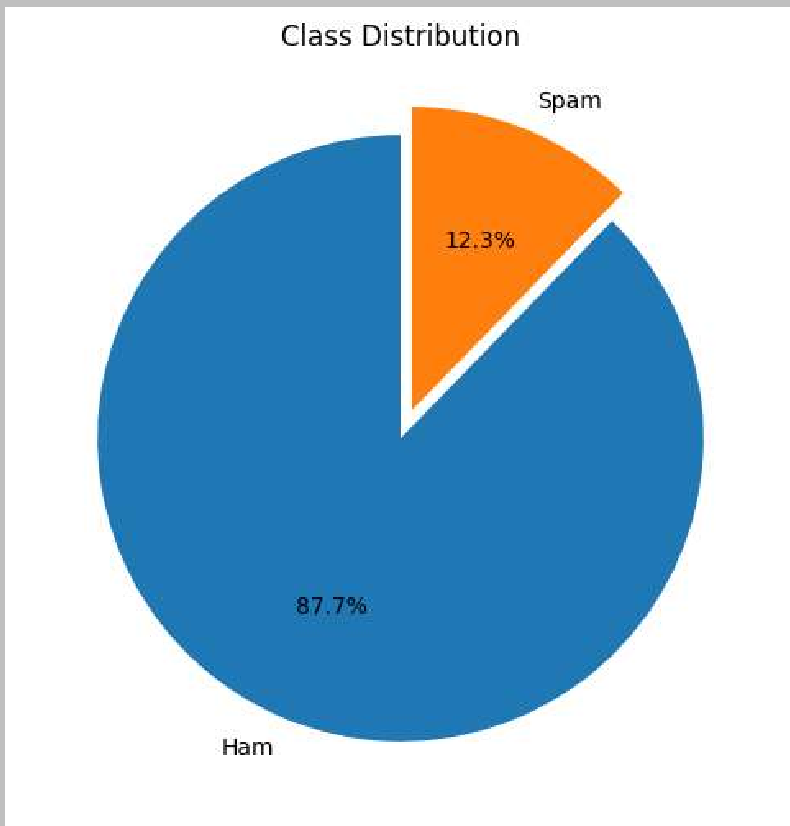
Deployment

- Proof-of-concept simulation: clients aggregate counts, IID assumption noted
- prsentation

CRISP-DM

Iterative loops between phases (e.g., evaluation feedback improved data preparation and feature choices

Data Overview



Synthetic_sms.csv (5700, 2)

Category	Count
Ham	5000
Spam	700
Total	5700

- 84% exact duplicates”
- 897 unique messages after deduplication
- Use hash groups to keep duplicates in same fold/client

Data Overview

Why Naïve Bayes Works

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Spam

Words: (190 Unique)
free, cash, call, win, claim, R1000, 37100

Bigrams: co za, visit http, voucher worth

Digit ratio: 0.11

URL presence: 32%

Alphabetic ratio: 0.69

Punctuation ratio: 0.04

35 words overlap

- Avg length (chars): ham 47.3, spam 88.7

Ham

Words:(233 unique)
you, have, meeting, good, thanks, love

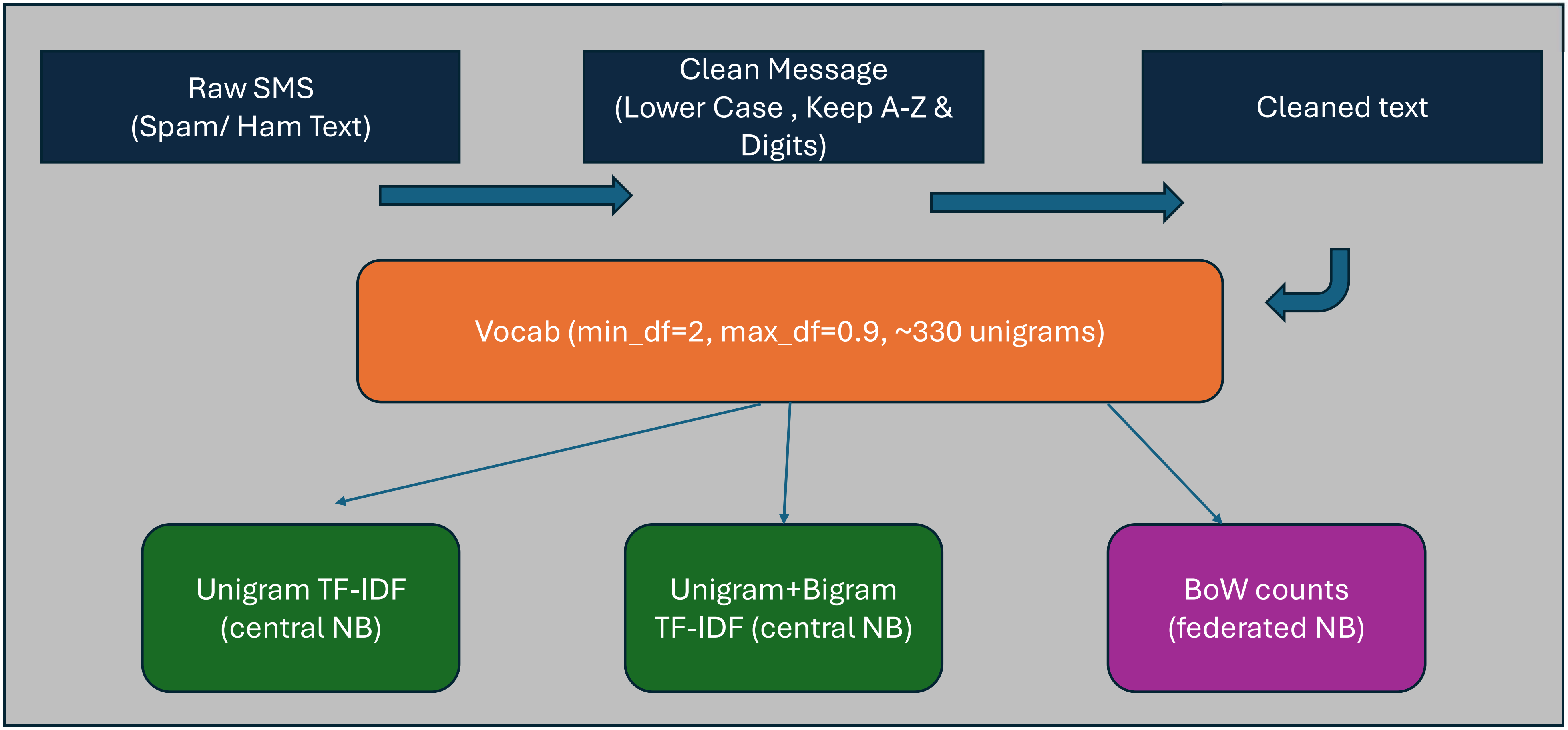
Digit ratio: 0.02

URL presence: 0%

Alphabetic ratio: 0.79

Punctuation ratio: 0.03

Data Preparation and Feature Representations



Centralized VS Federated Pipelines

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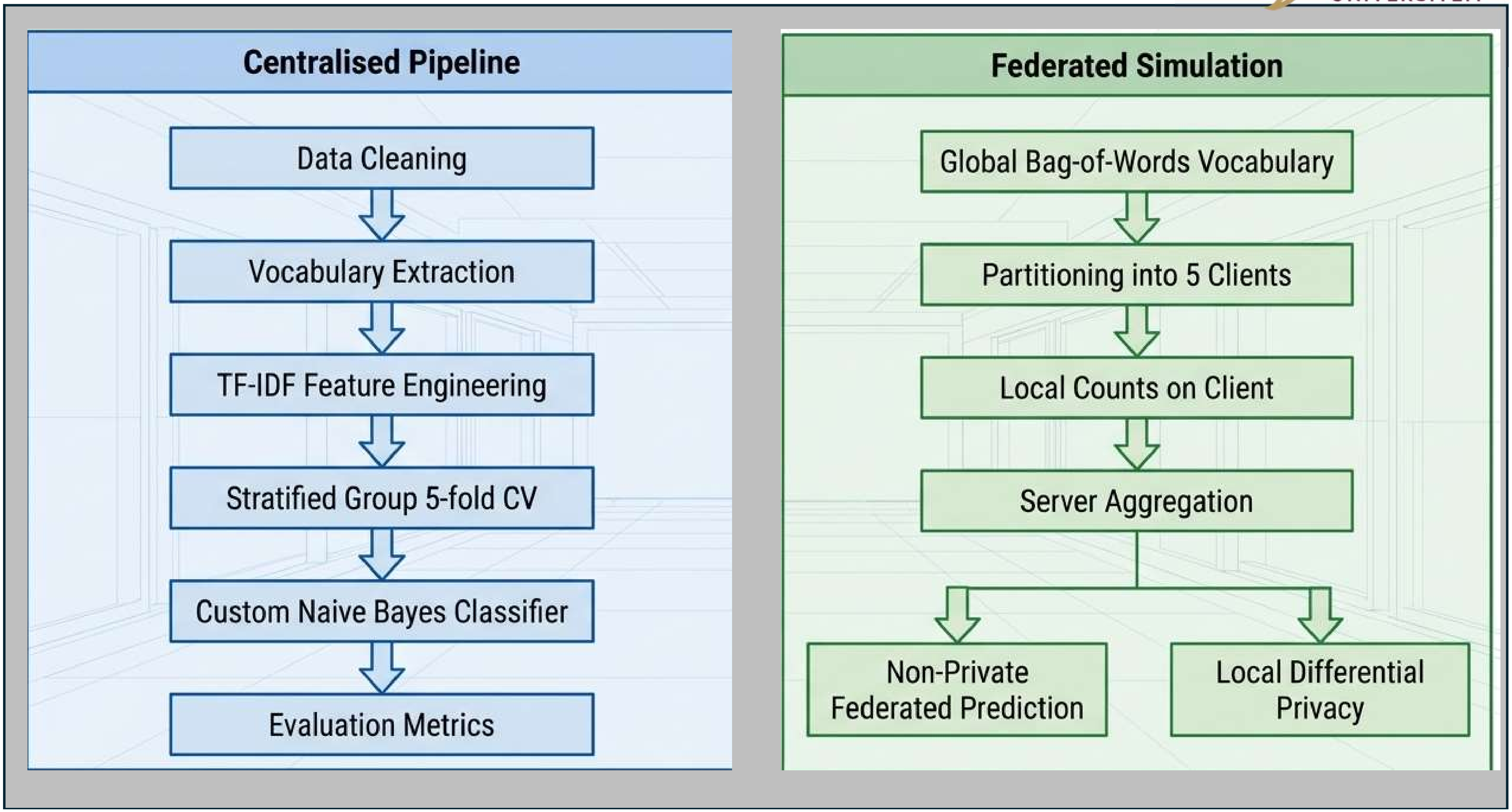
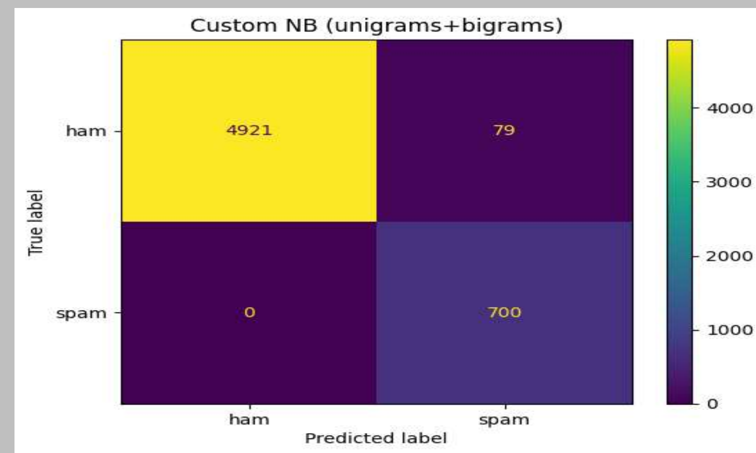
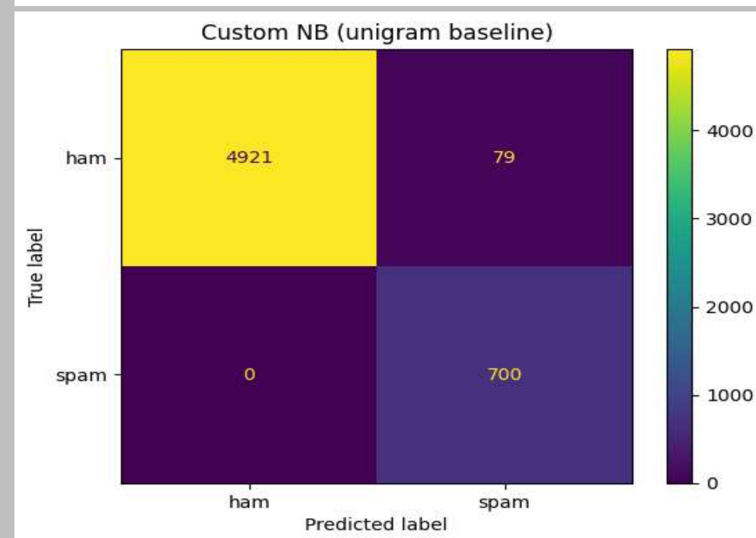


Modeling: Custom Naïve Bayes

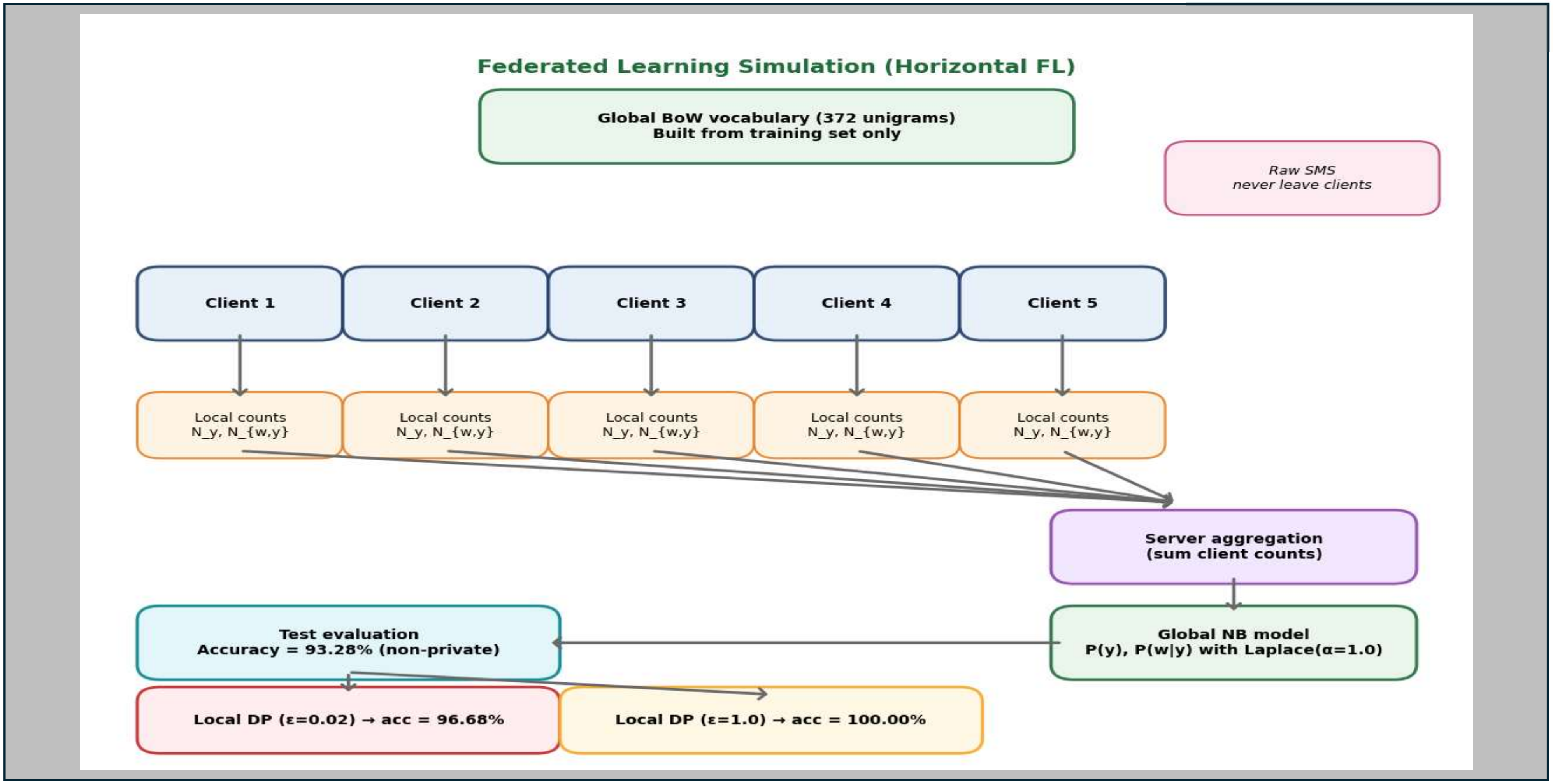
Metric	Unigram TF-IDF & Uni + Bigram TF-IDF
Accuracy	0.984
Precision	0.9206
Recall	1.0000
F1-Score	0.9537

- Custom NB (Laplace smoothing, log-probs)
- No false negatives
- 79 ham → spam (false positives)





Federated Learning Architecture: Client–Server Design



Federated vs Central Bag-of-Words Results & Feasibility Conclusion

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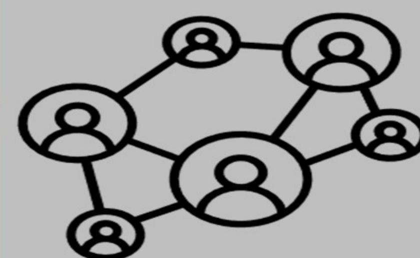
Model	Accuracy
Central BoW NB	0.9328
Federated BoW NB	0.9328

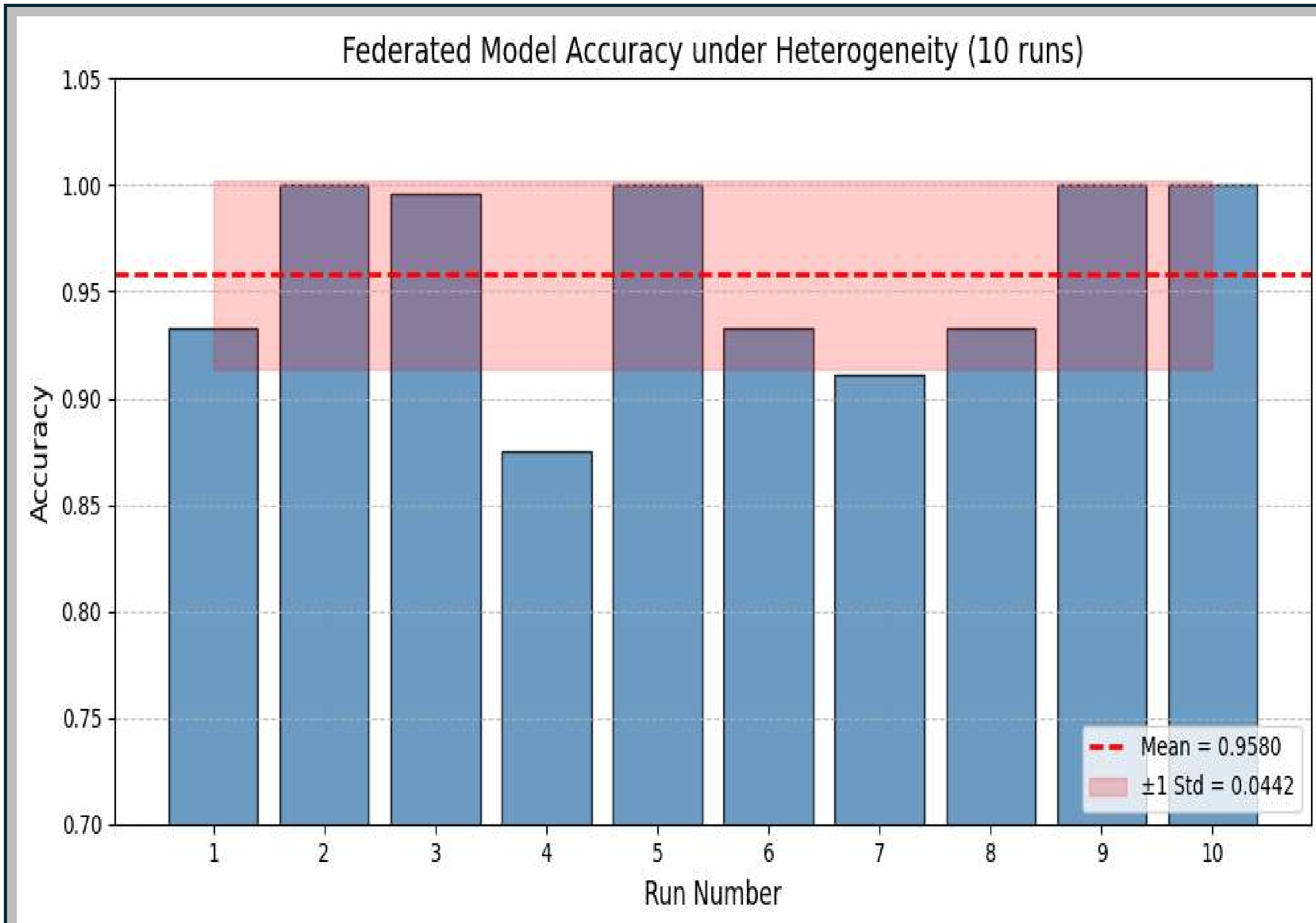
- Federated aggregation reproduces central NB exactly.

- Confirms Naïve Bayes is aggregatable via counts..



- NB is feasible for FL – additive counts + identical performance
- Federated accuracy varies slightly across runs (e.g., 93.28% in one run, but always matches the central BoW model)

Federated Simulation:
Robustness under Client Dropout and Capacity Limits



2/5 clients active; extreme capacity limits (some use 1% data)

10 random runs: mean **0.958 ± 0.044**.

Range 0.846–1.000 – variability but overall strong.

Differential Privacy: Privacy-Utility Trade-Off

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Model	Accuracy	Precision	Recall	F1
Non-private federated	0.9328	0.6343	1.0000	0.7762
DP ε=1.0	1.0000	1.0000	1.0000	1.0000
DP ε=0.02	0.9668	0.7988	0.9562	0.8704

ε ↓
(Stronger Privacy
+More Noise)

Accuracy drop due to
more ham-> Spam
(FP)

Conclusion & Future Work

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Key Findings

- Central TF-IDF NB: ≈0.99 acc, 1.0 recall (spam)
- Federated BoW NB ≈ central BoW NB → feasible
- DP ε controls privacy-utility trade-off
- Robust to some client heterogeneity

Limitations

- Synthetic, strongly separable dataset
- Simulated clients, single round, IID assumption (non-IID not tested)
- No secure aggregation / real-world deployment constraints
  No defence against malicious clients
  Fixed vocabulary (no adaptation to new words)

- Multi-round Federated Averaging (FedAvg) – McMahan et al. (2017)
- Lightweight transformers (e.g., DistilBERT) in federated setting (Lin et al, 2022)
- Secure aggregation

THANK YOU